

CodeBuddy Local

Offline AI error explainer and fix assistant for Snapdragon-powered HP PCs
Snapdragon AI Lab Build & Present Challenge | Participant: G N Bharath

Problem

Students often code with unreliable internet, and pasting coursework or project code into cloud AI tools raises privacy and cost concerns. Beginners also lose hours to cryptic error messages and stack traces they cannot interpret.

Solution

CodeBuddy Local is a lightweight assistant that runs entirely on the laptop. The user pastes a code snippet and its error message into a simple local web interface. A small code language model running on-device analyzes the input and returns a plain-English explanation of what went wrong, followed by a suggested fix. A beginner mode rewrites the explanation in simpler terms with short examples.

Key features

- Explains errors and stack traces in plain English
- Suggests corrected code with a short reason for each change
- Beginner mode for simpler, step-by-step explanations
- Works 100% offline after setup, so no code leaves the device

AI models and Snapdragon optimization

The solution uses a small open-source code LLM (Qwen2.5-Coder or Phi class), sourced from Qualcomm AI Hub or open-source platforms, and is optimized to run on Snapdragon-powered HP PCs. Inference is accelerated on the NPU through ONNX Runtime with the QNN execution provider. The project includes a benchmark comparing CPU and NPU on time to first token, tokens per second, and total time per fix.

Technology stack

Python backend, lightweight HTML/JavaScript front end, ONNX Runtime (QNN), Qualcomm AI Hub models.

Impact and future scope

CodeBuddy Local gives students a free, private, always-available coding helper that makes good use of Snapdragon on-device AI and low power consumption. Future work includes a VS Code extension for one-click fixes, support for more programming languages, and explanations for failing unit tests.